MATH 162, Winter 2026
SCRIPT 10: Compactness

Definition 10.1. We say that a function $f\colon A \to \mathbb{R}$ is *bounded* if $f(A)$ is a bounded subset of $\mathbb{R}$. We say that f is bounded above if $f(A)$ is bounded above and that f is bounded below if $f(A)$ is bounded below.

If $f\colon A \to \mathbb{R}$ is bounded above we say that f *attains* its least upper bound if there is some $a \in A$ such that $f(a) = \sup f(A)$. Similarly, if $f\colon A \to \mathbb{R}$ is bounded below we say that f *attains* its greatest lower bound if there is some $a \in A$ such that $f(a) = \inf f(A)$.

Exercise 10.2. If possible, find examples of each of the following; a picture suffices.

a) A continuous function on $[1, \infty)$ that is not bounded above.

b) A continuous function on $[1, \infty)$ that is bounded above but does not attain its least upper bound.

c) A continuous function on $(0, 1)$ that is not bounded below.

d) A continuous function on $(0, 1)$ that is bounded below but does not attain its greatest lower bound.

Proof. We will provide an example of each of the following examples:

(a) Consider the function $f(x) = x$.
On the domain $[1, \infty)$, we have that the domain goes to infinity and therefore, as $f(x)$ equals the input, when x gets larger the $f(x)$ gets larger without limit. Therefore, the function is not bounded above.

(b) Consider the function $f(x) = -\frac{1}{x}$.
Proof it is bounded above: For any $x \geq 1$, the value $\frac{1}{x}$ is positive. Therefore, $-\frac{1}{x}$ is always negative (less than 0). Since $f(x) < 0$ for all x, 0 is an upper bound.Proof the Sup is not attained: The Supremum (Least Upper Bound) is 0 because the values get arbitrarily close to 0 as x increases.

For the bound to be attained, there must be a point a in the domain where $f(a) = 0$. If we solve $-\frac{1}{a} = 0$, we get $-1 = 0$, which is impossible. Since no such a exists, the function never touches its bound.

(c) Consider the function $f(x) = -\frac{1}{x}$.
Suppose there is a lower bound L.Pick an x that is extremely small and positive, specifically $x = \frac{1}{|L|+1}$.This x is between 0 and 1, so it is in our domain.Calculating the function value: $f(x) = -(|L| + 1)$.This value is definitely lower than L.Since we found a point lower than the "floor," L cannot be a lower bound.

(d) Consider the function $f(x) = x$.
The range of this function on the domain $(0, 1)$ is the interval $(0, 1)$. The greatest lower bound (infimum) is 0. However, since $0 \notin (0, 1)$, the function never takes the value 0. Thus, the infimum is not attained. Proof it is bounded below: The domain is $(0, 1)$, which means every x we put in is strictly greater than 0. Since $f(x) > 0$ always, 0 is a lower bound.Proof the Inf is not attained: The Infimum (Greatest Lower Bound) is 0. For the bound to be attained, there must be a point a inside the set $(0, 1)$ where $f(a) = 0$. The only number that makes $f(x) = 0$ is 0 itself. However, 0 is not included in the open interval $(0, 1)$. Therefore, the function never reaches the value 0 inside this domain

□

We will see below that if the domain of a continuous function is *both* closed and bounded, then its image is bounded and attains its bounds. First, we introduce a new concept: compactness.

Definition 10.3. Let X be a subset of $\mathbb{R}$ and let $\mathcal{G} = \{G_\lambda\}_{\lambda\in\Lambda}$ be a collection of subsets of $\mathbb{R}$. We say that $\mathcal{G}$ is a *cover* of X if every point of X is in some G_λ, or in other words:

$$X \subset \bigcup_{\lambda\in\Lambda} G_\lambda.$$

We say that the collection $\mathcal{G}$ is an *open cover* if each G_λ is open.

Definition 10.4. Let X be a subset of $\mathbb{R}$. X is *compact* if for every open cover $\mathcal{G}$ of X, there exists a finite subset $\mathcal{G}' \subset \mathcal{G}$ that is also an open cover.

A good summary of the definition of compactness is "every open cover contains a finite subcover".

Exercise 10.5. Show that all finite subsets of $\mathbb{R}$ are compact.

Proof. Let S be a finite subset of $\mathbb{R}$ and let $\mathcal{G}$ be an arbitrary open cover of S. Since S is finite, we can list its elements as $S = \{s_1, s_2, \ldots, s_n\}$. By the definition of a cover, for each element $s_i \in S$, there must exist at least one open set $\mathcal{G}_i \in M$ such that $s_i \in \mathcal{G}_i$.

We define a new collection $\mathcal{G}' = \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_n\}$. This collection is a subcollection of M and is clearly finite as it contains at most n sets. Since every $s_i \in S$ is contained in $\mathcal{G}_i$, the union of the sets in $\mathcal{G}'$ contains S. Thus, $\mathcal{G}'$ is a finite subcover of S. Since every open cover of S has a finite subcover, S is compact. □

Lemma 10.6. *No finite collection of regions covers* $\mathbb{R}$.

Proof of Lemma 10.6. We want to show that no finite collection of regions covers $\mathbb{R}$. We will prove this through contradiction.

Assume for the sake of contradiction that $\mathcal{G}$ is a finite collection of open regions. Then, we can define $\mathcal{G}$ as

$$\mathcal{G} = \{G_1, G_2.G_3, \ldots, G_n\} \text{ for some } n \in \mathbb{N}$$

By the definition of an open region, we have that all regions G_i are defined as (a_i, b_i) for some $i \in \mathbb{N}$ where $a < b$. Now, consider

$$M = \max\{b_1, b_2, \ldots, b_n\}$$

which is the largest possible endpoint within the collection of regions. Then, consider the element $x = M + 1$. By the definition of $\mathbb{R}$, we know that $x \in \mathbb{R}$, however, as M represents the largest possible endpoint of the collection $\mathcal{G}$, we have that $x \notin \mathcal{G}$. Therefore, we have that $\mathbb{R} \not\subseteq \mathcal{G}$. This shows that $\mathcal{G}$ does not cover $\mathbb{R}$ and is a contradiction. As we have taken an arbitrary value of $n \in \mathbb{N}$, this works to show that no finite collection of regions covers $\mathbb{R}$. □

Theorem 10.7. $\mathbb{R}$ *is not compact.*

Proof. To show $\mathbb{R}$ is not compact, we want to show an open cover that possesses no finite subcover. We will prove by counterexample. Consider the collection $G = \{(x-1, x+1) \mid x \in \mathbb{R}\}$. This is an open cover of $\mathbb{R}$ because every $x \in \mathbb{R}$ is contained in its respective interval.

By Lemma 10.6, no finite collection of regions can cover $\mathbb{R}$. Since every element of G is a region, any finite subcollection of G is a finite collection of regions and therefore cannot cover $\mathbb{R}$. Since there is an open cover with no finite subcover, $\mathbb{R}$ is not compact. □

Exercise 10.8. Show that regions are not compact.

Exercise 10.8. We want to show that regions are not compact. To show that regions are not compact, it suffices to show that there exists an open cover with no finite subcover of the region.

Let (a, b) be an arbitrary region. Now, consider the collection of regions $\mathcal{G} = \{(a + \frac{b-a}{n}, b) \mid n \in \mathbb{N}\}$. By definition, we have that $(a, b) \subset \mathcal{G}$. Within this construction, we have that as n increases, it creates a region G_λ that is greater than the previous region and therefore covers more of (a, b).

Suppose we took a finite subcover of $\mathcal{G}$. Let the finite subcover be defined as $\mathcal{G}' = \{G_1, G_2, \ldots, G_n\}$. Let $L = \min\{a+\frac{b-a}{1}, a+\frac{b-a}{2}, \ldots, a+\frac{b-a}{n}\}$. Now consider $x = a+\frac{b-a}{n+1}$. Then, we have that $x \in (a, b)$ but $x \notin \mathcal{G}$. Hence, no finite subcover and regiosn are not compact.

□

Theorem 10.9. *If X is compact, then X is bounded. Moreover, if X is also nonempty, then* $\sup X$ *and* $\inf X$ *exist and are contained in X.*

Proof of Theorem 10.9. Let X be compact. We want to prove that X is bounded.

To show that X is bounded below, consider the collection $\mathcal{G} = \{(-n, \infty) \mid n \in \mathbb{N}\}$. Since $\mathbb{N}$ is unbounded above, we have that $X \subset \bigcup \mathcal{G}$, and therefore $\mathcal{G}$ is an open cover of X.

Since X is compact, there exists a finite subcover $\mathcal{G}'$ of $\mathcal{G}$. We can write $\mathcal{G}' = \{(-n, \infty) \mid n \in A\}$ where $A \subset \mathbb{N}$ is a finite set. Let $N = \max A$. Then, we have that $\bigcup \mathcal{G}' = (-N, \infty)$, and therefore $X \subset (-N, \infty)$, so X is bounded below by $-N$. Symmetric reasoning shows that X is bounded above, and therefore X is bounded.

Now, let $X \neq \emptyset$. We want to show that $\sup X$ and $\inf X$ exist and are contained in X. Since X is bounded, $\inf X$ and $\sup X$ exist.

Then, assume for the sake of contradiction that $\inf X \notin X$. Consider the collection $\mathcal{G} = \{(\inf X + \frac{1}{n}, \infty) \mid n \in \mathbb{N}\}$. Since $\mathbb{N}$ is unbounded above, for any $x \in X$ we have that $x > \inf X$, and therefore there exists some $n \in \mathbb{N}$ such that $x > \inf X + \frac{1}{n}$, which means $x \in (\inf X + \frac{1}{n}, \infty)$. Therefore $\mathcal{G}$ is an open cover of X.

Since X is compact, there exists a finite subcover $\mathcal{G}' = \{(\inf X + \frac{1}{n}, \infty) \mid n \in B\}$ where $B \subset \mathbb{N}$ is a finite set. Let $N = \min B$. Then, we have that $\bigcup \mathcal{G}' = (\inf X + \frac{1}{N}, \infty)$, and therefore $X \subset (\inf X + \frac{1}{N}, \infty)$. But this means that $\inf X + \frac{1}{N}$ is a lower bound for X that is strictly greater than $\inf X$, contradicting the definition of $\inf X$.

Therefore $\inf X \in X$. Symmetric reasoning shows that $\sup X \in X$, and this suffices to complete the proof. $\square$

We can now discuss the relationship between continuity and compactness. We will then turn to showing how compact sets are related to closed, bounded sets.

Theorem 10.10. *Suppose that $f\colon X \to \mathbb{R}$ is continuous. If X is compact, then $f(X)$ is compact.*

Theorem 10.10. Suppose that $f : X \to \mathbb{R}$ is continuous. We want to show that if X is compact then $f(X)$ is compact.

Let X be an arbitrary compact set. Therefore, by **Definition 10.4**, we know that for all open covers M of X, there exists a finite subset $M' \subset M$ that is also an open cover. Hence, by the definition of X, we may say there exists an open cover of X. Now, suppose there exists an infinite collection of open sets U such that $U = \{G_\lambda\}_{\lambda \in \Lambda}$ where G_λ is an open set within the cover of $f(X)$. To prove that $f(X)$ is compact, it suffices to show that every open cover of $f(X)$ has a finite subcover

As f is continous, by **Definition 9.4**, For each $G_\lambda \in U$, the preimage $f^{-1}(G_\lambda)$ is open. Then, we get that there exists an infinite open collection of covers on X. Then, as X is compact, we may say that for every open cover G_λ of X , there exists a finite subset $G'_\lambda \subset G_\lambda$. Now, we may construct a collection of finite subsets $U' = \{G_1, G_2, G_3, \dots, G_n\}$ that suffices to cover all $x \in X$. Therefore, we have that X is contained entirely in the collection of finite subsets.

Since X is compact, we have found a finite collection of preimages $\{f^{-1}(G_1), f^{-1}(G_2), \dots, f^{-1}(G_n)\}$ that covers X. This means:

$$X \subset f^{-1}(G_1) \cup f^{-1}(G_2) \cup \ldots \cup f^{-1}(G_n)$$

Now, we apply the function f to both sides of this inclusion. As f maps the domain to the codomain, we have:

$$f(X) \subset f\left(f^{-1}(G_1) \cup f^{-1}(G_2) \cup \ldots \cup f^{-1}(G_n)\right)$$

By the properties of images and unions, the image of a union is the union of the images. Furthermore, we know from our earlier work that $f(f^{-1}(G_i)) \subset G_i$. Therefore:

$$f(X) \subset f(f^{-1}(G_1)) \cup \ldots \cup f(f^{-1}(G_n)) \subset G_1 \cup \ldots \cup G_n$$

This shows that $f(X)$ is contained entirely in the union of the finite sub-collection $\{G_1, G_2, \dots, G_n\}$. Since we have successfully reduced an arbitrary open cover of $f(X)$ to a finite subcover, we conclude that $f(X)$ is compact. $\square$

Corollary 10.11. *If $X \subset \mathbb{R}$ is non-empty, compact and $f\colon X \longrightarrow \mathbb{R}$ is continuous, then $f(X)$ has a first and last point.*

Proof of Corollary 10.11. Suppose $X \subset \mathbb{R}$ is non-empty, compact, and $f : X \to \mathbb{R}$ is continuous. We want to show that $f(X)$ has a first and last point. For this, it suffices to show that the supremum and infimum are contained in $f(X)$

As $X \subset \mathbb{R}$ is compact and f is continuous, we have that $f(X)$ is compact. Then, by **Theorem 10.9**, we have that $f(X)$ is bounded. Furthermore, as $f(X)$ is nonempty, then $\sup X$ and $\inf X$ exists and are contained in X.

Now, the supremum and infimum exist and are contained in X. Let $s = \sup f(X)$ and $i = \inf f(X)$. Therefore, by definition, there exists no point $f(x) \in f(X)$ such that $f(x) > s$ and $f(X) < i$. Furthermore, $s, i \in f(X)$. Hence, this suffices to prove that $f(X)$ contains a first and a last point. $\square$

We now give a characterization of compact sets. We have already seen (in Theorem 10.9) that they are bounded. We will now prove that boundedness together with closedness are defining characteristics. This is in Theorem 10.19; first we need some preliminary results

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Lemma 10.12. *Let $X \subset \mathbb{R}$ and $p \in \mathbb{R} \setminus X$. Then*

$$\mathcal{G} = \{\text{ext}(\underline{ab}) \mid p \in \underline{ab}\}$$

is an open cover of X.

Proof of Lemma 10.12. Suppose that $X \subset \mathbb{R}$ and $p \in \mathbb{R} \setminus X$. We want to show that $\mathcal{G} = \{\text{ext}(\underline{ab}) \mid p \in \underline{ab}\}$.

Let $x \in X$. By definition, we know that $x \neq p$. Therefore, we may construct an interval (a, b) such that $a < p < b$ and $x \notin (a, b)$. Then, we may say that $x \in \text{ext}(\underline{ab})$. As $x \in \text{ext}(\underline{ab})$, we have that $x \in \mathcal{G}$.

As we have taken an arbitrary $x \in X$, we may say that $\forall x \in X, x \in \mathcal{G}$. Therefore, as $\mathcal{G}$ is a collection of open subsets of $\mathbb{R}$, as it is a collection of $\text{ext}(\underline{ab})$, we have that $\mathcal{G}$ is an open cover of $\mathbb{R}$. □

Theorem 10.13. *If X is compact, then X is closed.*

Proof of Theorem 10.13. We want to prove that if X is compact then X is closed. For this, it suffices to show that $\mathbb{R} \setminus X$ is open.

Suppose for the sake of a contradiction that X is not closed. Then, there exists $p \in LP(X)$ such that $p \notin X$, then $p \in \mathbb{R} \setminus X$.

Then, $\mathcal{G} = \{\text{ext}((a, b)) \mid p \in (a, b)\}$ is an open cover of X by Lemma 10.12. Since X is compact, there exists a finite subcover $\mathcal{G}'$.

Then, $\mathcal{G}' = \{\text{ext}(\underline{a_1, b_1}), \text{ext}(\underline{a_2, b_2}), \ldots, \text{ext}(\underline{a_n, b_n})\}$. Let V be the intersection of the corresponding open intervals:

$$V = (\underline{a_1, b_1}) \cap (\underline{a_2, b_2}) \cap \cdots \cap (\underline{a_n, b_n})$$

Since V is a finite intersection of open intervals containing p, V is itself an open interval containing p.

Since $X \subset \bigcup_{G \in \mathcal{G}'} G$, it follows by De Morgan's Law that X is contained in the complement of the intersection of the intervals. That is, $X \subset V^c$. This implies that $V \cap X = \emptyset$.

Consequently, V is a neighborhood of p such that $V \cap (X \setminus \{p\}) = \emptyset$. This contradicts the assumption that p is a limit point of X, as every neighborhood of a limit point must contain at least one point of X other than p. Therefore, p must be in X, and X is closed.

This completes the proof.

□

For the next three results, fix points $a, b \in \mathbb{R}$ and suppose $\mathcal{G}$ is an open cover of $[a, b]$.

Lemma 10.14. *For all $s \in [a, b]$, there exist $G \in \mathcal{G}$ and $p, q \in \mathbb{R}$ such that $p < s < q$ and $[p, q] \subset G$.*

Proof of Lemma 10.14. Let $[a, b]$ be an open cover. Then, there exists an open cover $\mathcal{G}$ that covers $[a, b]$. Then, by definition, $\forall s \in [a, b]$, there exists some $G \in \mathcal{G}$ such that $s \in G$. We want to show that there exists some $p, q \in \mathbb{R}$ such that $p < s < q$ where $[p, q] \subset G$.

Consider an arbitrary $s \in [a, b]$. By the definition of a cover, we have that there exists some $G_\lambda \in \mathcal{G}$ such that $s \in G_\lambda$. As G_λ is open, we have that there $\exists \delta > 0$ such taht $(s - \delta, s + \delta) \subset G$. Now, let $p = s - \frac{\delta}{2}$ and $q = s + \frac{\delta}{2}$. By defintion we have that $q > p$ and that $p < s < q$. Furthermore, we can construct an interval $[p, q]$ such that $[p, q] = \{p \leq x \leq q\}$ for some $x \in \mathbb{R}$.

Then, we have that there exist some $p, q \in \mathbb{R}$ such taht $p < s < q$ and $[p, q] \subset G$. □

Lemma 10.15. *Let X be the set of all $x \in \mathbb{R}$ that are* reachable from a*, by which we mean the following: there exist $n \in \mathbb{N} \cup \{0\}$, $x_0, \ldots, x_n \in \mathbb{R}$ and $G_1, \ldots, G_n \in \mathcal{G}$ such that $a = x_0 < x_1 < \ldots < x_{n-1} < x_n = x$ and $[x_{i-1}, x_i] \subset G_i$ for $i = 1, \ldots, n$.*

(Note in particular that $a \in X$, by choosing $n = 0$.)

Then the point b is not an upper bound for X.

(Hint: suppose b is an upper bound, and apply Lemmas 10.14 and 5.10 to $s = \sup X$.)

Proof of Lemma 10.15. Suppose for the sake of contradiction that b is an upper bound for X. We know that $a \in X$ and hence $X \neq \emptyset$. Therefore, as X is nonempty and bounded above, we have that $s = \sup X$ exists.

Then, since b is an upper bound, we have that $s \leq b$, and therefore $s \in [a, b]$. By **Lemma 10.14**, there exist $G \in \mathcal{G}$ and $p, q \in \mathbb{R}$ such that $p < s < q$ and $[p, q] \subset G$. Since $p < s = \sup X$, we have that p is not an upper bound of X, and therefore there exists $x \in X$ such that $p < x \leq s$.
Since $x \in X$, there exist $x_0, \ldots, x_n \in \mathbb{R}$ and $G_1, \ldots, G_n \in \mathcal{G}$ such that $a = x_0 < x_1 < \ldots < x_n = x$ and $[x_{i-1}, x_i] \subset G_i$ for all i. Now, let $G_{n+1} = G$ and $x_{n+1} = q$. Then, we have that

$$[x_n, x_{n+1}] = [x, q] \subset [p, q] \subset G = G_{n+1}.$$

Therefore, we have that q is reachable from a, and hence $q \in X$. But as $q > s$, this contradicts the assumption that $s = \sup X$ is an upper bound of X.
Therefore, b is not an upper bound for X, which suffices to complete the proof. □

Theorem 10.16. *The set $[a, b]$ is compact.*

Proof of Theorem 10.16. We want to show that set $[a,b]$ is compact. For that it suffcies to show that for any arbitrary open cover $\mathcal{G}$ there exists a finite number of subcovers to represent $[a,b]$.

Let $[a,b]$ be a finite, closed interval. By **Lemma 10.15**, we have that there exists $n \in \mathbb{N} \cup \{0\}$ such that $a = x_0$ and $x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n$ where $[x_{i-1}, x_i] \subset G$.

As $[a,b]$ is closed, we may write it as $\bigcup [x_{i-1}, x_i]_{i \in |n|} \subset [a,b]$ such that $s \in [x_{i-1}, x_i]$ and $s \in G$. Then, as $i \in |n|$, we have that there exists a finite number of $G \in \mathcal{G}$. Hence, $[a,b]$ is compact. □

Corollary 10.17. *If $f\colon [a,b] \longrightarrow \mathbb{R}$ is continuous, then there exists a point $c \in [a,b]$ such that $f(c) \geq f(x)$ for all $x \in [a,b]$. Similarly, there exists a point $d \in [a,b]$ such that $f(d) \leq f(x)$ for all $x \in [a,b]$.*

Proof of Corollary 10.17. Firstly, we want to show that if $f : [a,b] \to \mathbb{R}$ is continuous, then there exists a point $c \in [a,b]$ such that $f(c) \geq f(x)$ for all $x \in [a,b]$.

By definition of $[a,b]$, we may say $[a,b] \subset \mathbb{R}$ and $[a,b]$ is nonempty. Furthermore, by **Theorem 10.16**, we have that $[a,b]$ is compact.

Then, as f is continuous, $[a,b] \subset \mathbb{R}$, and by $[a,b]$ being compact and nonempty, we may say $f([a,b]$ has a first and last point. Therefore, we may suffices there exists some $c \in [a,b]$ such that $f(c) \geq f(x), \forall x \in [a,b]$. Similarly, there must exists a point $d \in [a,b]$ such that $f(d) \leq f(x), \forall x \in [a,b]$.

This suffices to prove the statement. □

Theorem 10.16 is a key ingredient for the proof of the Heine-Borel theorem. We need just one more theorem first.

Theorem 10.18. *A closed subset Y of a compact set $X \subset \mathbb{R}$ is compact.*

(Question: Does it matter whether Y is closed in X or in $\mathbb{R}$?)

Proof of Theorem 10.18. We want to show that a closed subset Y of a compact set $X \subset \mathbb{R}$ is compact. To show that Y is compact, it suffices to show that every open cover of Y has a finite subcover.

Let $\{U_\alpha\}$ be an open cover of Y. Since Y is closed, we have that $\mathbb{R} \setminus Y$ is open. Therefore, $\{U_\alpha\} \cup \{\mathbb{R} \setminus Y\}$ is an open cover of X, as every $x \in X$ is either in Y and hence covered by some U_α, or in $X \setminus Y \subset \mathbb{R} \setminus Y$.

Since X is compact, there exists a finite subcover of X, so that $X \subset U_{\alpha_1} \cup U_{\alpha_2} \cup \cdots \cup U_{\alpha_n} \cup (\mathbb{R} \setminus Y)$ for some finite collection $U_{\alpha_1}, \ldots, U_{\alpha_n}$. Since $Y \subset X$ and $Y \cap (\mathbb{R} \setminus Y) = \emptyset$, we have that $Y \subset U_{\alpha_1} \cup U_{\alpha_2} \cup \cdots \cup U_{\alpha_n}$. Therefore, Y has a finite subcover and hence is compact.

Furthermore, it does not matter whether Y is closed in X or in $\mathbb{R}$. Since X is compact, X is closed in $\mathbb{R}$, and therefore a set that is closed in X is also closed in $\mathbb{R}$. Hence the two notions coincide here and the proof holds either way. $\square$

Theorem 10.19. *Let $X \subset \mathbb{R}$. X is compact if and only if X is closed and bounded.*

Proof of Theorem 10.19. Let $X \subset \mathbb{R}$. We want to show that X is compact if and only if X is closed and bounded.

($\Rightarrow$) Let X be compact. We want to show that X is closed and bounded. By **Theorem 10.13**, we have that X is closed. Furthermore, by **Theorem 10.9** we have that X is bounded. Therefore, we have that if X is compact, then X is closed and bounded.

($\Leftarrow$) Let X be closed and bounded. We want to show that X is compact. Since X is bounded, there exist $a, b \in \mathbb{R}$ such that $X \subset [a, b]$. By **Theorem 10.16**, we have that $[a, b]$ is compact. Then, by **Theorem 10.18**, as X is a closed subset of the compact set $[a, b]$, we have that X is also compact.

This suffices to prove that X is compact if and only if X is closed and bounded. $\square$

As a consequence we obtain a version of a theorem dating back to Bolzano in 1817 (though our proof is very different). First, though, a lemma which can be proved straight from the definition of compactness.

Lemma 10.20. *A compact set $X \subset \mathbb{R}$ with no limit points must be finite.*

Proof of Lemma 10.20. Let X be a compact subset of $\mathbb{R}$ such that X has no limit points. We want to show that X is finite.

Since X has no limit points, we have that for all $x \in X$, there exists a region R_x such that $x \in R_x$ and $X \cap R_x \setminus \{x\} = \emptyset$. Now, consider the collection $\mathcal{G} = \{R_x\}_{x \in X}$. As each R_x is a region and therefore open, we have that $\mathcal{G}$ is an open cover of X.

Since X is compact, there exists a finite subcover $\mathcal{G}'$. We can write $\mathcal{G}' = \{R_{x_1}, R_{x_2}, \ldots, R_{x_n}\}$ for some $n \in \mathbb{N}$.

Now, we have that for all $i \in \{1, \ldots, n\}$, $R_{x_i} \cap X \setminus \{x_i\} = \emptyset$, and therefore x_i is the only element of X contained in R_{x_i}.

Therefore, we have that $|X| = |\mathcal{G}'| = n$, which is finite. This suffices to prove that X is finite. $\square$

Theorem 10.21. *Every bounded infinite subset of $\mathbb{R}$ has at least one limit point.*

Proof of Theorem 10.21. We want to show that every bounded infinite subset of $\mathbb{R}$ has at least one limit point. We note that the contrapositive of **Lemma 10.20** says that if $M \subset \mathbb{R}$ is infinite and compact, then it has at least one limit point.

Let X be a bounded infinite subset of $\mathbb{R}$. We claim that $\overline{X}$ is closed, bounded, and infinite. We have that $\overline{X}$ is closed by definition. To see that $\overline{X}$ is bounded, let a be a lower bound for X and let b be an upper bound for X. Then, we have that $X \subset [a, b]$, and therefore $\overline{X} \subset \overline{[a, b]} = [a, b]$ since $[a, b]$ is closed. Hence, $\overline{X}$ is bounded. To see that $\overline{X}$ is infinite, we have that $\overline{X} = X \cup LP(X) \supset X$, and since X is infinite, it follows that $\overline{X}$ is infinite.

Therefore, we have that $\overline{X}$ is closed, bounded, and infinite, and hence compact by **Theorem 10.19**. Then, by the contrapositive of **Lemma 10.20**, we have that $LP(\overline{X}) \neq \emptyset$.

Now, we have that

$$\emptyset \neq LP(\overline{X}) = LP(X \cup LP(X)) = LP(X) \cup LP(LP(X)) \subset LP(X)$$

where we use that $LP(LP(X)) \subset LP(X)$ and that limit points distribute over finite unions. Therefore, we have that $LP(X) \neq \emptyset$, which suffices to prove that every bounded infinite subset of $\mathbb{R}$ has at least one limit point. □

Additional Exercises

1. Prove that a set $X \subset \mathbb{R}$ is compact if, and only if, every open cover made up only of regions has a finite subcover.

2. For each of the following sets give an open cover that has no finite subcover. Justify your answers.

   (a) $\{\frac{1}{n} \mid n \in \mathbb{N}\}$

   (b) $\{x \in \mathbb{Q} \mid x \leq 2\}$.

3. (a) Suppose that X and Y are compact subsets of $\mathbb{R}$. Show that $X \cup Y$ is compact and $X \cap Y$ are compact.

   (b) If X_1, X_2, … are compact sets, are the following compact:

$$\bigcap_{n=1}^{\infty} X_n \quad \text{and/or} \quad \bigcup_{n=1}^{\infty} X_n?$$

Additional Exercise 3(a). Let X and Y be compact subsets of $\mathbb{R}$. Firstly, we want to show that $X \cup Y$ is compact. For this it suffices to show that $X \cup Y$ is closed and bounded.

We have that X and Y are compact and therefore it follows that X and Y are closed and bounded. Furthermore, as X and Y are closed, they contain all of their limit points.

Now consider $X \cup Y$. By **Corollary 4.18**, a finite union of closed sets is closed, and hence it follows that $X \cup Y$ is closed. To see that $X \cup Y$ is bounded, let a_X be the lower bound of X and a_Y be the lower bound of Y, and let b_X be the upper bound of X and b_Y be the upper bound of Y. Then for any $x \in X \cup Y$, we have $\min(a_X, a_Y) \leq x \leq \max(b_X, b_Y)$, so $X \cup Y$ is bounded.

Therefore, we have that $X \cup Y$ is closed and bounded, and thus compact.

Secondly, we want to show that $X \cap Y$ is compact. By **Theorem 4.15**, an arbitrary intersection of closed sets is closed, and so $X \cap Y$ is closed. To see that $X \cap Y$ is bounded, note that $X \cap Y \subset X$, and since X is bounded, every element of $X \cap Y$ is also an element of X and is therefore bounded by the same bound as X. Therefore $X \cap Y$ is closed and bounded, and hence compact. $\square$

Additional Exercise 3(b). Let $X_1, X_2, \ldots$ be compact subsets of $\mathbb{R}$. Since each X_n is compact, it follows that each X_n is both closed and bounded. We consider the infinite intersection and infinite union separately.

Firstly, we want to show that $\bigcap_{n=1}^{\infty} X_n$ is compact. To show that $\bigcap_{n=1}^{\infty} X_n$ is compact, it suffices to show that it is closed and bounded.

To show that $\bigcap_{n=1}^{\infty} X_n$ is closed, we have that each X_n is compact and therefore closed. Therefore, **Theorem 4.15**, an arbitrary intersection of closed sets is closed, and hence it follows that $\bigcap_{n=1}^{\infty} X_n$ is closed.

Then, we want to show that $\bigcap_{n=1}^{\infty} X_n$ is bounded. We have that $\bigcap_{n=1}^{\infty} X_n \subset X_1$, and since X_1 is compact it is bounded. Therefore, as every element of $\bigcap_{n=1}^{\infty} X_n$ is also an element of X_1, it follows that $\bigcap_{n=1}^{\infty} X_n$ is bounded by the same bound as X_1.

Therefore, we have that $\bigcap_{n=1}^{\infty} X_n$ is closed and bounded, and thus compact.

Secondly, we want to show that $\bigcup_{n=1}^{\infty} X_n$ need not be compact. We prove through counterexample.

Consider the sets $X_n = [n, n+1]$ for each $n \in \mathbb{N}$. Each X_n is a closed and bounded interval and therefore compact. However, we have that

$$\bigcup_{n=1}^{\infty} X_n = [1, \infty),$$

which is unbounded as it contains arbitrarily large elements, and therefore not compact. Hence, an infinite union of compact sets need not be compact. $\square$

4. (a) Let $X_1, X_2, X_3 \cdots$, be nonempty compact subsets of $\mathbb{R}$ such that $X_1 \supset X_2 \supset X_3 \supset \cdots$. Prove that

$$\bigcap_{i=1}^{\infty} X_i \neq \emptyset.$$

Hint: Proof by contradiction.

(b) Consider a sequence of nonempty closed intervals $I_k \subset \mathbb{R}, k \in \mathbb{N}$ such that $I_{k+1} \subset I_k$ and $|I_k| < \frac{1}{k}$, for all k. (Here $|I_k|$ denotes the length of the interval.) Prove that the intersection

$$\bigcap_{k=1}^{\infty} I_k$$

consists of exactly one point.

Additional Exercise 4(a). Let $X_1, X_2, X_3, \ldots$ be nonempty compact subsets of $\mathbb{R}$ such that $X_1 \supset X_2 \supset X_3 \supset \cdots$. We want to show that $\bigcap_{i=1}^{\infty} X_i \neq \emptyset$.

Assume for the sake of contradiction that $\bigcap_{i=1}^{\infty} X_i = \emptyset$. Since each X_n is compact, each X_n is closed, and therefore each complement $U_i = \mathbb{R} \setminus X_i$ is open. Since $\bigcap_{i=1}^{\infty} X_i = \emptyset$, by De Morgan's law we have that $\bigcup_{i=1}^{\infty} U_i = \mathbb{R}$, and therefore $\{U_i\}_{i=1}^{\infty}$ is an open cover of X_1.

Since X_1 is compact, there exists a finite subcover, so that $X_1 \subset U_1 \cup U_2 \cup \cdots \cup U_N$ for some $N \in \mathbb{N}$. Now, since the X_i are decreasing, that is $X_1 \supset X_2 \supset X_3 \supset \cdots$, it follows that the complements $U_i = \mathbb{R} \setminus X_i$ are increasing, that is $U_1 \subset U_2 \subset U_3 \subset \cdots$. Therefore we have that $U_1 \cup U_2 \cup \cdots \cup U_N = U_N = \mathbb{R} \setminus X_N$, and so $X_1 \subset \mathbb{R} \setminus X_N$, which gives us $X_1 \cap X_N = \emptyset$.

However since $X_N \subset X_1$ and X_N is nonempty, we have that $X_1 \cap X_N = X_N \neq \emptyset$, a contradiction.

Therefore it must be that $\bigcap_{i=1}^{\infty} X_i \neq \emptyset$. $\square$

Additional Exercise 4(b). Let $I_k \subset \mathbb{R}$ be a sequence of nonempty closed intervals such that $I_{k+1} \subset I_k$ and $|I_k| < \frac{1}{k}$ for all $k \in \mathbb{N}$. We want to show that $\bigcap_{k=1}^{\infty} I_k$ consists of exactly one point.

Firstly, we want to show that $\bigcap_{k=1}^{\infty} I_k$ contains at least one point. Since each I_k is a nonempty closed and bounded interval in $\mathbb{R}$, each I_k is compact. Furthermore, we have that $I_{k+1} \subset I_k$ for all k, so the intervals are nested and decreasing. Therefore, by **Additional Exercise 4(a)**, it follows that $\bigcap_{k=1}^{\infty} I_k \neq \emptyset$, and hence the intersection contains at least one point.

Secondly, we want to show that $\bigcap_{k=1}^{\infty} I_k$ contains at most one point. Suppose for the sake of contradiction that there exist two distinct points $x, y \in \bigcap_{k=1}^{\infty} I_k$ with $x \neq y$. Then for every $k \in \mathbb{N}$, both x and y lie in I_k, and therefore

$$|x - y| \leq |I_k| < \frac{1}{k}.$$

Since this holds for all $k \in \mathbb{N}$, taking $k \to \infty$ gives $|x - y| \leq 0$, and since $|x - y| \geq 0$ we get $|x - y| = 0$, which means $x = y$. This contradicts our assumption that $x \neq y$.

Therefore $\bigcap_{k=1}^{\infty} I_k$ consists of exactly one point. $\square$

5. Prove that $\mathbb{R}$ is uncountable.

Hint: Suppose that $f : \mathbb{N} \longrightarrow \mathbb{R}$ is a bijection. You need to find a point in $\mathbb{R}$ that is not in $f(\mathbb{N})$. To do this, start by recursively defining closed intervals X_i such that $X_i \subset X_{i-1}$ and $f(i) \notin X_i$ and use Problem 4.

Additional Exercise 5. Assume for the sake of contradiction that $\mathbb{R}$ is countable. Then there exists a bijection $f : \mathbb{N} \longrightarrow \mathbb{R}$, so that every real number appears in the list $f(1), f(2), f(3), \ldots$

We recursively construct a sequence of nonempty closed intervals $X_1 \supset X_2 \supset X_3 \supset \cdots$ such that $f(i) \notin X_i$ and $|X_i| < \frac{1}{i}$ for all $i \in \mathbb{N}$. We construct these intervals as follows.

For the base case, take any closed interval $X_1 \subset \mathbb{R}$ such that $f(1) \notin X_1$ and $|X_1| < 1$. This is always possible since $f(1)$ is just a single point and we can always find a closed interval in $\mathbb{R}$ that avoids it.

Now suppose we have constructed X_{i-1}. Since X_{i-1} is a closed interval containing infinitely many points and $f(i)$ is just a single point, we can always find a closed subinterval $X_i \subset X_{i-1}$ such that $f(i) \notin X_i$ and $|X_i| < \frac{1}{i}$. To see this, if $f(i) \notin X_{i-1}$ then we simply take any closed subinterval of X_{i-1} with length less than $\frac{1}{i}$. If $f(i) \in X_{i-1}$, then $f(i)$ splits X_{i-1} into two closed subintervals, at least one of which has length less than $\frac{1}{i}$, and we take X_i to be that subinterval.

Now we have a sequence of nonempty closed intervals $X_1 \supset X_2 \supset X_3 \supset \cdots$ with $|X_i| < \frac{1}{i}$ for all $i \in \mathbb{N}$. By **Additional Exercise 4(b)**, the intersection $\bigcap_{i=1}^{\infty} X_i$ consists of exactly one point; call it x^*.

Since f is a bijection onto $\mathbb{R}$, there exists some $n \in \mathbb{N}$ such that $f(n) = x'$. But by construction, $f(n) \notin X_n$, and since $x' \in \bigcap_{i=1}^{\infty} X_i \subset X_n$, we have that $x' \in X_n$. This is a contradiction.

Therefore $\mathbb{R}$ is uncountable. $\square$

6.

Definition 10.22. Define the set $\mathbb{R}_{\text{ext}}$ to be the set of *extended Dedekind cuts*. These are those sets which satisfy 6.1(b) and 6.1(c). In other words $\mathbb{R}_{\text{ext}} = \mathbb{R} \cup \{\emptyset, \mathbb{Q}\}$. We give $\mathbb{R}_{\text{ext}}$ the same order as $\mathbb{R}$. Namely, if $A, B \in \mathbb{R}_{\text{ext}}$ then $A < B$ if and only if $A \subsetneq B$.

(a) Which of Axioms 1-4 does $\mathbb{R}_{\text{ext}}$ satisfy?

Definition 10.23. Any subset $\mathcal{R} \subset \mathbb{R}_{\text{ext}}$ is a region if:

i. $\mathcal{R} = \underline{AB}$, or

ii. $\mathcal{R} = \{E \in \mathbb{R}_{\text{ext}} | E < A\}$, or

iii. $\mathcal{R} = \{E \in \mathbb{R}_{\text{ext}} | E > A\}$,

for some $A, B \in \mathbb{R}$.

(b) Show that a set $\mathcal{V} \subset \mathbb{R}_{\text{ext}}$ is open if and only if for every $A \in \mathcal{V}$, there exists a region $\mathcal{R} \subset \mathcal{V}$ such that $A \in \mathcal{R}$.

(c) Suppose that $\mathcal{E} \subset \mathbb{R}_{\text{ext}}$ is bounded away from $\emptyset$ and $\mathbb{Q}$ in the sense that, there exists $A, B \in \mathbb{R}$ such that $A \leq E \leq B$ for all $E \in \mathcal{E}$.

i. Show that $\emptyset, \mathbb{Q} \notin \mathcal{E}$.

ii. Show that if $\mathcal{V} \subset \mathbb{R}_{\text{ext}}$ is open, then there exists a an open set $\mathcal{W} \subset \mathbb{R}$ such that $\mathcal{E} \cap \mathcal{V} = \mathcal{E} \cap \mathcal{W}$.

(d) Show that $\mathbb{R}_{\text{ext}}$ is compact. What do $\emptyset$ and $\mathbb{Q}$ represent in terms of concepts you have seen before?